The Privacy-Protected Gridded Environmental Impacts Frame

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Abstract

This paper introduces the Gridded Environmental Impacts Frame (Gridded EIF), a novel privacy-protected dataset derived from the U.S. Census Bureau's confidential Environmental Impacts Frame (EIF) microdata infrastructure. The EIF combines comprehensive administrative records and survey data on the U.S. population with high-resolution geospatial information on environmental hazards. While access to the EIF is restricted due to the confidential nature of the underlying data, the Gridded EIF offers a broader research community the opportunity to glean insights from the data while preserving confidentiality. We describe the data and privacy protection process, and offer guidance on appropriate usage, presenting practical applications.

* Corresponding Author: John Voorheis, john.l.voorheis@census.gov. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Data Management System (DMS) number: P-7505723, Disclosure Review Board (DRB) approval numbers: CBDRB-FY24-0394 and CBDRB-FY24-CES023-019). We thank Surya Menon and Yolande Tra for research assistance. Colmer is grateful to the National Science Foundation (CAREER Award SES:2338220), Sloan Foundation, Russell Sage Foundation, Washington Center for Equitable Growth, Upjohn Institute, and the University of Virginia's Bankard Fund for Political Economy and Center for Real Estate and the Built Environment for generous funding and support, which has contributed to the development of the underlying Environmental Impacts Frame microdata.

1 Introduction

Advances in the availability of new microdata infrastructures, such as the Census Environmental Impacts Frame (EIF), provide unprecedented opportunities for researchers to expand our understanding of how environmental conditions are distributed across individuals with diverse demographic and economic characteristics, and to study the causes and consequences of environmental exposures (Voorheis et al., 2023). However, due to the highly confidential nature of the data, access is only available to researchers who qualify for and pass necessary background checks, and whose projects have undergone a lengthy adn thorough approval processes. These barriers to entry unavoidably slow the process of research and consequently delay the production of new insights. To address this challenge we introduce the Gridded Environmental Impacts Frame—a publicly accessible data product which sensibly aggregates the EIF microdata in ways that preserve privacy while maintaining consistency with the confidential microdata. The creation of this data allows for expanded data access to institutions and researchers who might otherwise lack resources or the necessary conditions for approval to access the confidential EIF microdata.

The Gridded EIF provides a publicly available, privacy protected complement to the confidential EIF microdata and presents opportunities for researchers to conduct rich spatial analyses while protecting individual identities. The data is organized on a fixed 0.01-degree grid (approximately 1.2km^2 in North America) and provides counts of the population by age, sex, and race and ethnicity as well as counts of the population by race and ethnicity and household income decile.

Providing the data in this form allows researchers to take advantage of certain features of the EIF, offering substantial benefits over traditional analyses of aggregated place-based data. First, the data enables distributional analysis by intersectional characteristics, such as by race *and* income. Second, the underlying administrative records are updated more frequently than are most aggregate demographic data, which means that the most recent EIF data is more timely. For instance, as of May 2024, roughly two thirds of the U.S.

population has appeared on a 1040 for tax year 2023 (filed in early 2024), and close to the entire population is covered in the 2023 vintage of the EIF's residential history file. This is in contrast to the most recently available ACS 5 year tables, which draw from survey responses as far back as 2018. Third, most individuals in the EIF can be geocoded to precise latitude and longitude, allowing for flexible aggregation to any geographic unit, not limited to just those that are defined by the Census Bureau (such as Censustracts). This is particularly important when analyzing hazards that do not align with administrative boundaries—in these cases, aggregating to units of fixed size (such as geographic grid) as opposed to fixed population (such as Census tracts) may enhance analysis. By developing the Gridded EIF as a public-facing product that preserves intersectional information, that is available on a fixed geographic grid, and that has higher-frequency temporal information, thus extending the analytical capabilities of the confidential EIF to a broader set of researchers and stakeholders.

The paper lays out the process for building the Gridded EIF, discusses the privacy-protection processes that we use, illustrates the consequences of these privacy-protection processes for inference in the data by drawing comparisons with the confidential microdata, and provides an illustrations of how the data could be used.

2 Building the Gridded Environmental Impacts Frame

The Gridded EIF is an aggregation of the EIF microdata infrastructure. The EIF microdata is built as a modular infrastructure, allowing researchers to construct custom datasets suited to each specific type of analysis. The three key components of the EIF are: 1) a residential history module tracking individual residential locations for nearly the entire U.S. population from 1999 to 2023, 2) a demographic spine containing basic demographic information (race, ethnicity, sex, place of birth, date of birth and mortality) for all individuals with social security numbers, sourced from the Social Security Administration's Numident file, and 3) an environmental hazards module, which harmonizes geospatial information on

environmental hazards such as air pollution and wildfires, allowing for an easy link to the residential locations in the RHF. The EIF infrastructure is based on administrative records, which may not always fully capture all subpopulations. As such, the population totals in the EIF microdata and the gridded EIF may deviate from population totals in the Decennial Census or Census Bureau population estimates. The EIF's modular design facilitates the integration of additional data sources, such as the Opportunity Databank, consisting of harmonized administrative tax data, or commercially provided housing data available in the Census Bureau, creating a breadth of research opportunities.

The Gridded EIF is constructed directly from the demographic spine and residential history modules of the Environmental Impacts Frame, which we combine with income information from the Opportunity Databank. Adjusted Gross Income (AGI) from IRS 1040s is aggregated to the household (Master Address File ID, MAFID) level using similar rules as those used in Kondo et al. (2024).[1] The Gridded EIF starts from this merged dataset in each year and aggregates the data in two ways. First, all geocoded individuals are assigned to a grid point on a fixed 0.01 degree grid. Each grid point is a fixed (unprojected) 0.01 degree square, which is about 1.2 km^2 in North America. The merged microdata is then collapsed by race, sex and age groups (under 18, 18-64, and 65+); and by race/ethnicity and income deciles, within these grid points. Consistent with the EIF we define 6 harmonized categories for race and ethnicity: Hispanic, Non-Hispanic White, Non-Hispanic Black, Non-Hispanic Asian, Non-Hispanic American Indian and Alaskan Native and Non-Hispanic Other. Individuals who cannot be matched to the demographic spine (e.g., individuals without social security numbers) will not have age and sex information; the race-by-age-by-sex aggregations retain counts of these individuals as a separate group in each grid point. The grid system used is essentially identical in the North American domain to the widely used unprojected

[1]For MAFIDs with fewer than 10 unique tax units, the household income is calculated as the sum of all unique tax-unit level AGI values in that MAFID. For MAFIDs with 10 or more unique tax units, we assume, consistent with the evidence in Kondo et al. (2024), that this is a data quality issue involving improper MAFID assignment in multiunit structures and assign the average tax unit level AGI as the household income for that MAFID.

grid systems in atmospheric sciences, for instance the satellite-derived PM2.5 data from van Donkelaar et al. (2021), facilitating the harmonization of external environmental data sources at the level of variation in socioeconomic characteristics.

2.1 Noise Injection Process

The grid points used represent very small geographies (a 0.01 degree grid point is about 1.2 km^2 in the North American domain, which is about the size of the median Census block group), which means it is not possible to publish these raw tabulations under current Census Bureau disclosure avoidance guidelines without additional privacy protection. To protect the privacy of individuals in the underlying microdata, in line with these rules, we use a noise infusion strategy inspired by the differential privacy methods used in the 2020 Decennial Census(Abowd et al., 2022). The noise injection method proceeds as follows. First, we inject mean zero, discrete Gaussian noise with standard deviation $\sigma = \frac{\gamma}{2\times\rho}$, where γ can be interpreted as the squared sensitivity of the statistics to be released and ρ is a parameter capturing the privacy loss budget.[2] In a differential privacy framework, these two parameters together describe the privacy guarantee of the noise infusion method (Canonne et al., 2020).

We select values for γ and ρ in consultation with the Census Bureau's Disclosure Review Board, which result in most draws from the resulting discrete Gaussian distribution to fall between -3 and 3.[3]. This noise can be thought of as "on the order of rounding" as it represents a similar amount of coarsening as the rounding schemes used for official Census tabulations of the American Community Survey and other collections. This noise is only infused into demographic counts that actually exist in the underlying microdata for each grid point. If a particular race-by-income group or race-by-sex-by-age group does not exist in the underlying microdata, the count of these individuals is treated as structurally zero. This approach to noise injection and the selection of ρ and γ parameters differ slightly from

[2]In a formal privacy setting, "sensitivity" represents the maximum amount that a single observation can affect a given statistic or set of statistics calculated from a given set of microdata.

[3]The parameters selected are $\rho \approx 0.33$ and $\gamma = 2$

strict differential privacy requirements, which would tune the parameters based on a measure of global sensitivity. Thus, our method is inspired by differential privacy methods, as in the methods used in Chetty and Friedman (2019), but lacks a formal privacy guarantee. The Census Bureau has judged that this lack of a formal guarantee is outweighed by the ability to release data products (such as the gridded EIF) which otherwise might not be feasible to release with a formal guarantee, and has judged the amount of privacy protection produced by the noise injection to meet the statutory guidelines of confidentiality under Title 13 of the U.S. Code.

The resulting datasets from this process consist of noisy integer counts by race-age-sex and race-income categories for each grid point. Because the discrete Gaussian noise can result in negative values that are in some cases larger in magnitude than the pre-noise counts in a given demographic group-by-grid point cell, the raw noisy counts are in practice sometimes negative. For many settings where the grid point level information is being aggregated to some larger geography, the noise will "net out", resulting in estimates consistent with the underlying microdata. However, in some settings, researchers will prefer demographic-by-grid point values to be strictly non-negative. To address this need, we implement a post-processing algorithm which leverages the nested structure of grids and the intuition that demographics and many types of environmental hazards are positively correlated within small geographic areas.

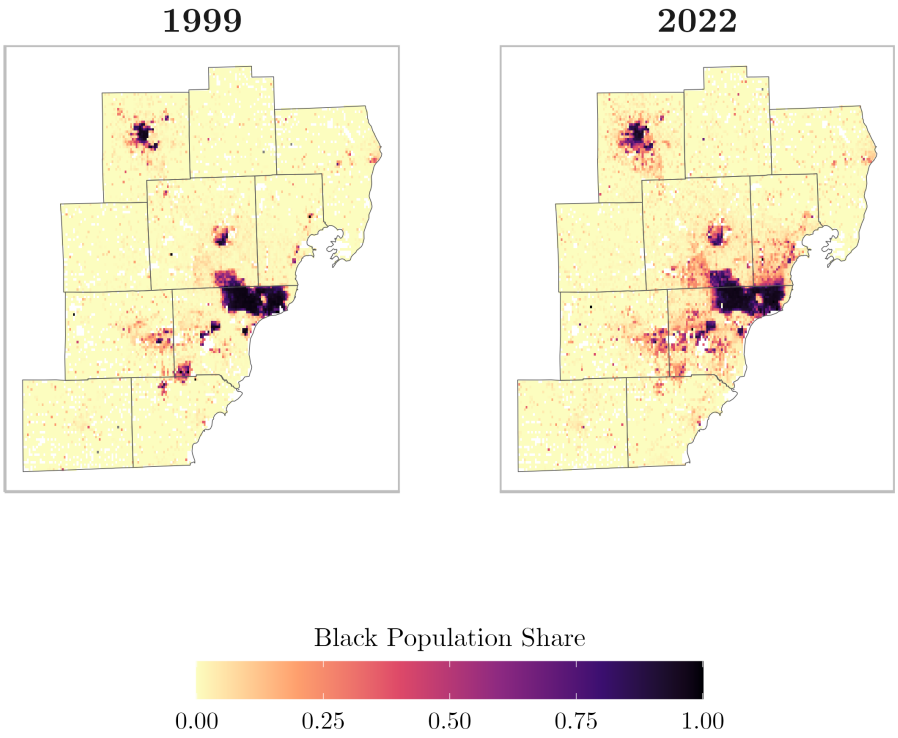
The post-processing algorithm proceeds as follows. First, "small noisy cells" are defined within 1 degree grid points as demographic cells with a raw noisy count smaller than the absolute value of the most negative value in that grid point (so for instance, if the most negative value is -5, then all cells in the 1 degree grid smaller than 5 would be classified as "small noisy cells"). The sum of all such noisy cells is then calculated by race group within each 1 degree grid. The value of all small noisy cells is then replaced with this sum divided by the number of small noisy cells within race group by grid point. By evenly dividing the total across all small noisy cells within race groups by 1 degree grid points for each dataset (race-

age-sex, race-by-income) this algorithm forces most cells to be non-negative, and ensures that population distributions by race are retained. Finally, if, after redistributing values across small noisy cells there remain some negative values, these are imputed to 0. Because this post-processing algorithm is evenly distributing small populations within 1 degree grid points (100 km $\times$ 100 km at the equator, about the size of Orange County, CA), it is better able to accurately capture racial differences in exposure within small areas, especially for smaller race groups, than the raw noisy measures, as long as the environmental hazards are similarly spatially correlated. For flexibility, both the raw noisy counts and the post-processing counts are provided. Thus, users will be able to apply a different post-processing algorithm if they wish, as long as the algorithm only uses information in the raw noisy counts.

2.2 Demographic Data

The gridded EIF population data alone offers opportunities to explore the changing demographics of the United States. By providing harmonized and consistent measures of population by race and income, it enables researchers to explore demographic shifts at temporal and geographic scales previously unavailable in public use data products. Figure 1 illustrates the power of this approach, by mapping the Black population share within grid points in the Detroit Combined Statistical Area in 1999 and 2022. The 1999 map reproduces the well-documented segregation of the metro area, with the predominantly Black city of Detroit sharply abutting majority white suburbs, especially to the north on the Eight Mile Rd. border. The gridded EIF's annual data allows for nuanced analysis of how this demographic pattern has evolved over time. Comparing the 1999 and 2022 maps illustrates the increasing suburbanization of the Black population in the Detroit Metro area, resulting in a marked reduction in the stark segregation between Detroit and its suburbs that was evident in 1999.

Figure 1: Black Population Share, Detroit Combined Statistical Area



Source: Gridded Environmental Impacts Frame. These maps show the Non-Hispanic Black population within each 0.01 degree grid point as a share of all individuals with non-missing race information, in 1999 and 2022. Opacity of each grid point is based on grid point total population, with lower population cells being more transparent.

2.3 Incorporating Spatially-Resolved Environmental Data

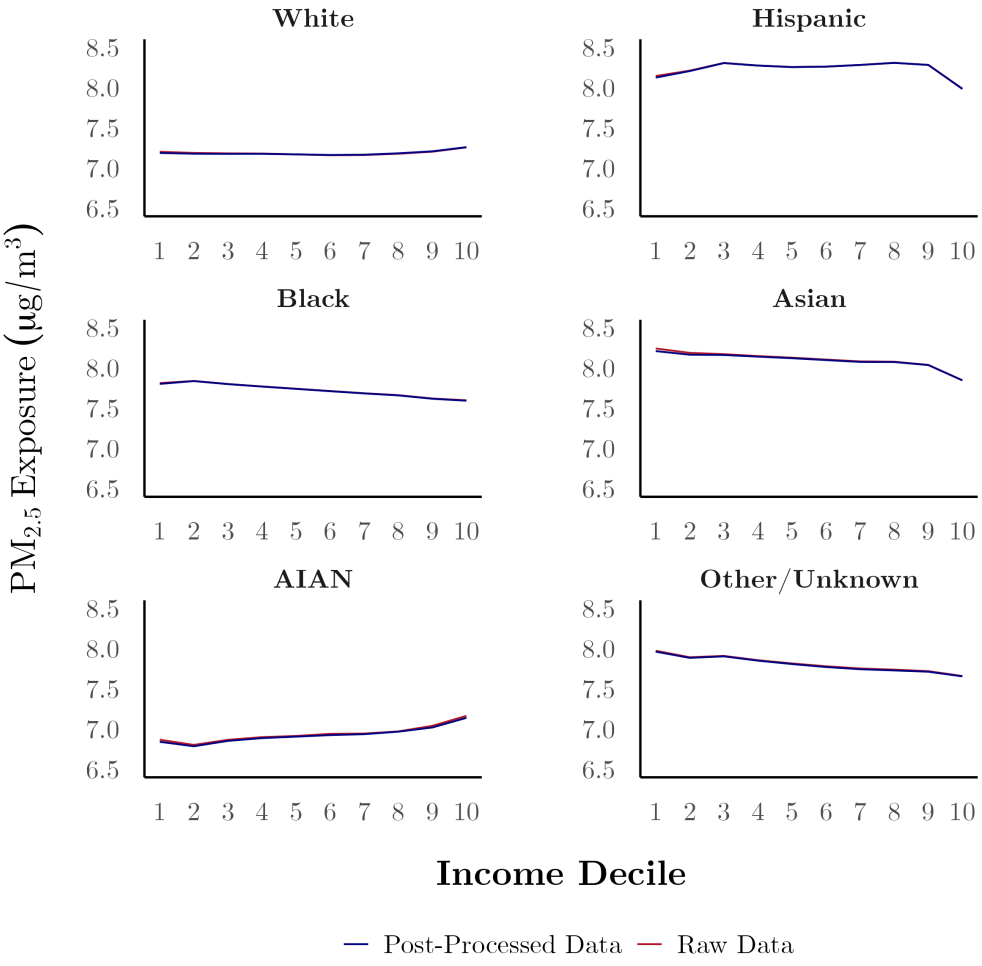
To aid researchers who are interested in exploring the distribution of exposure to environmental hazards, we have harmonized three commonly used geospatial datasets on air pollution to the same grid as the privacy protected demographic data described above. These data are: satellite derived PM2.5 from van Donkelaar et al. (2021) and Shen et al. (2024) provided by the Atmospheric Composition Analysis Group (ACAG), which span 1999-2022; satellite-derived PM2.5, NO2 and Ozone data from Requia et al. (2020), Di et al. (2020) and Di et al. (2019), provided by NASA's Socioeconomic Data and Applications Center (SEDAC), which span 2000-2016; and Land-use regression based estimates of PM2.5, PM10, SO2, NO2, CO and Ozone from Kim et al. (2020), made available by the Center for Air, Climate, and Energy Solutions (CACES), which spans 1999-2020. These harmonized gridded pollution data will be made available to researchers alongside the gridded population data.

To demonstrate the utility and quality of inference of the Gridded EIF at variousgeographic scales, we reproduce the core descriptive results presented in Colmer et al. (2024). Figure 2 presents race-specific income–PM2.5 exposure gradients for 2022 using the gridded EIF and the ACAG PM2.5 data.[4] We calculate these profiles using both the raw noisy values from each race-income-grid point and the noisy values after post-processing. For all race groups the two noisy measures produce very similar results. Figure 3 replicates this exercise for the Mankato, MN CZ, which has a population of around 100,000 and is a typical small metro area. At this smaller geographic scale, differences between the two noisy measures are more apparent. In particular, the raw noisy measure exhibits excess variability relative to the post-processing measure. Absolute differences, however, remain very small.

Repeating this analysis for each CZ, we calculate that the mean absolute error compared to the confidential microdata is about 65% larger when using the raw noisy measures than

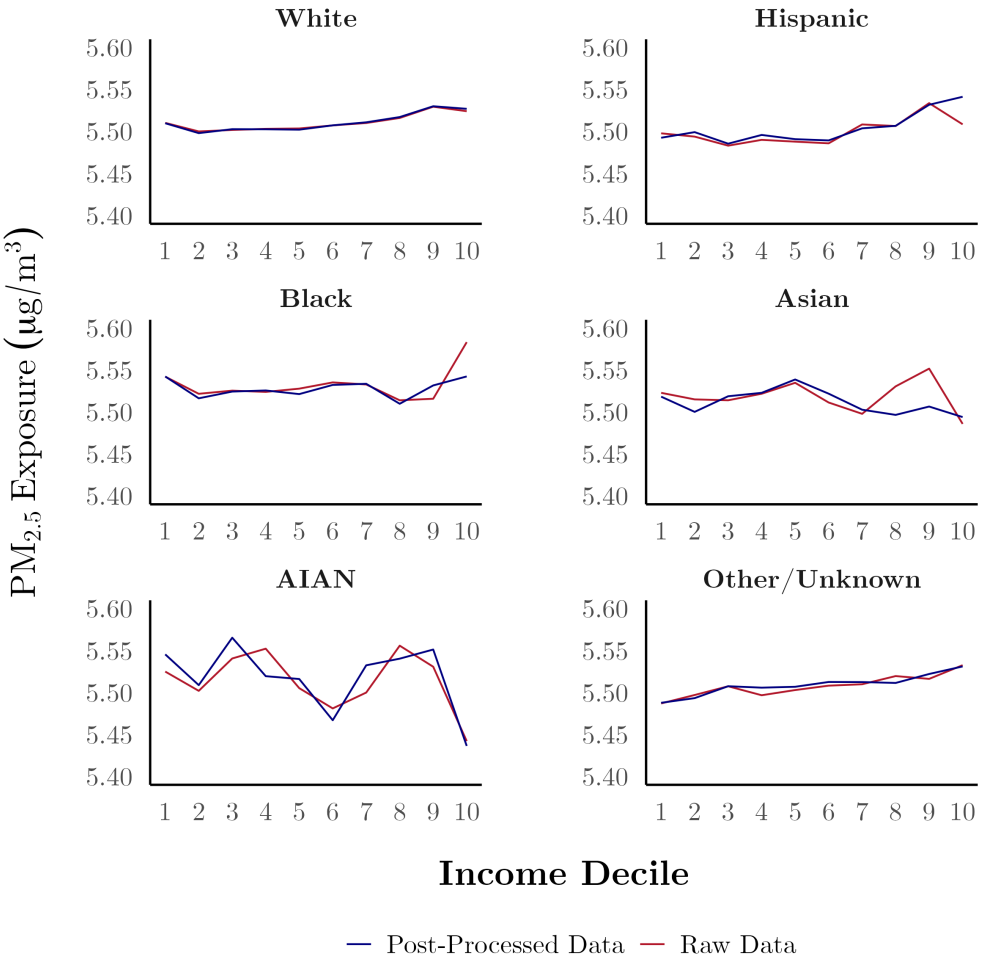
[4]We use the V5.GL.04 version of the ACAG data, which is referred to as *pm25_ACAG_gwr* in the gridded EIF data.

Figure 2: Race by Income PM2.5 Gradients, National



Source: Gridded Environmental Impacts Frame. This graph show the average PM2.5 exposure by race and income decile, for the nation as a whole, using the two noisy measures in the gridded EIF.

Figure 3: Race by Income PM2.5 Gradients, Mankato, MN



Source: Gridded Environmental Impacts Frame. This graph show the average PM2.5 exposure by race and income decile, for Mankato, MN, using the two noisy measures in the gridded EIF.

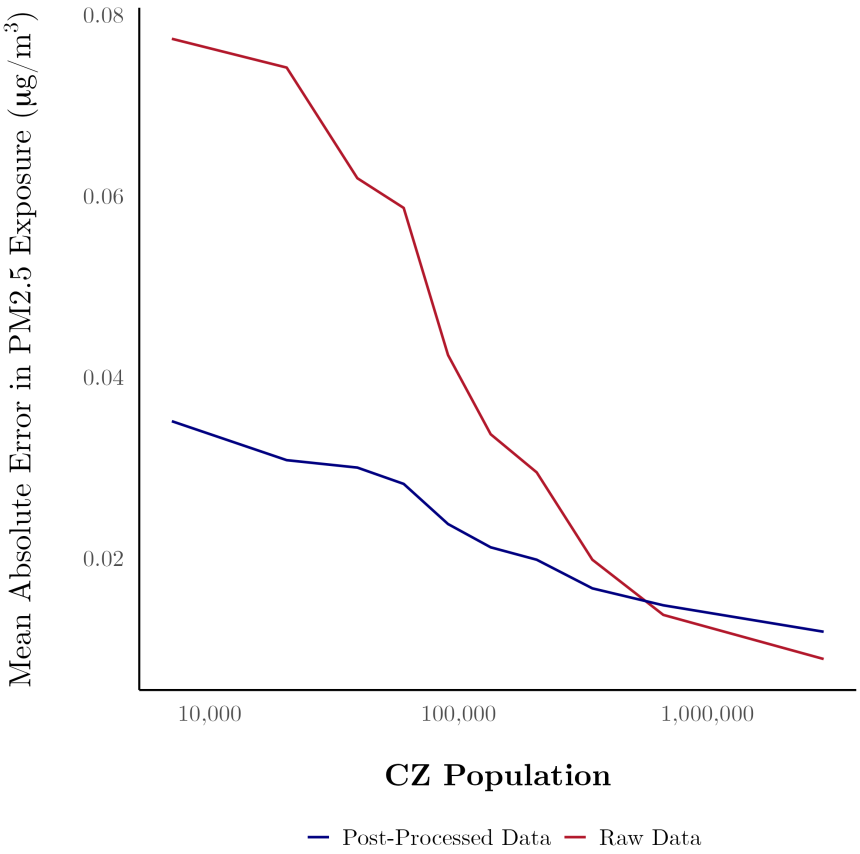
the mean absolute error when using the post-processing noisy measures. This varies by Commuting Zone (CZ) size, however, as shown in Figure 4. For the smallest CZs, defined as the bottom two quintiles by population, the raw noisy measures have around more than double the mean absolute error compared with the post-processing measures, while for larger CZs (the top 10% by population), the post-processing noisy measures produce about 25% larger mean absolute errors. The magnitudes of mean absolute error are small throughout the distribution. The crossover between these two mean absolute error curves happens right around the population threshold at which Census Bureau disclosure avoidance rules allow direct estimates from the confidential microdata (the population of the smallest state, Wyoming, which was around 600,000 in 2023).

3 "Real-Time" Analysis

In addition to the "standard" gridded EIF dataset produced using the 1999-2023 microdata files for the EIF, we also create a "real time" version of the EIF for 2024. This real time file starts from the most recent EIF residential histories, and then updates residential locations for all individuals who have filed their tax year 2023 1040 tax return through the spring of the current (2024 when this first version of the gridded EIF was created). This provides the most up-to-date information on individual's locations available in the administrative records underlying the EIF.

Using this updated version of the EIF, it is possible to analyze the demographics of populations exposed to environmental hazards in near real time. This could be particularly useful in the context of short term forecasting tools—the National Weather Service and other agencies have increasingly accurate forecasting capabilities for a number of weather and air quality related hazards. The gridded EIF provides information on the characteristics of the populations forecasted to be exposed to these hazards, making it possible to characterize

Figure 4: Mean Absolute Error in PM2.5 Exposure Estimates, Gridded EIF vs. Confidential Microdata



Source: Gridded Environmental Impacts Frame. This graph show the Mean Absolute Error between the actual CZ-level average PM2.5 exposure and the average PM2.5 exposure derived from the noisy measures in the gridded EIF, by CZ population.

Figure 5: The spatial distribution of hurricane-force wind speed probabilities, Hurricane Milton.

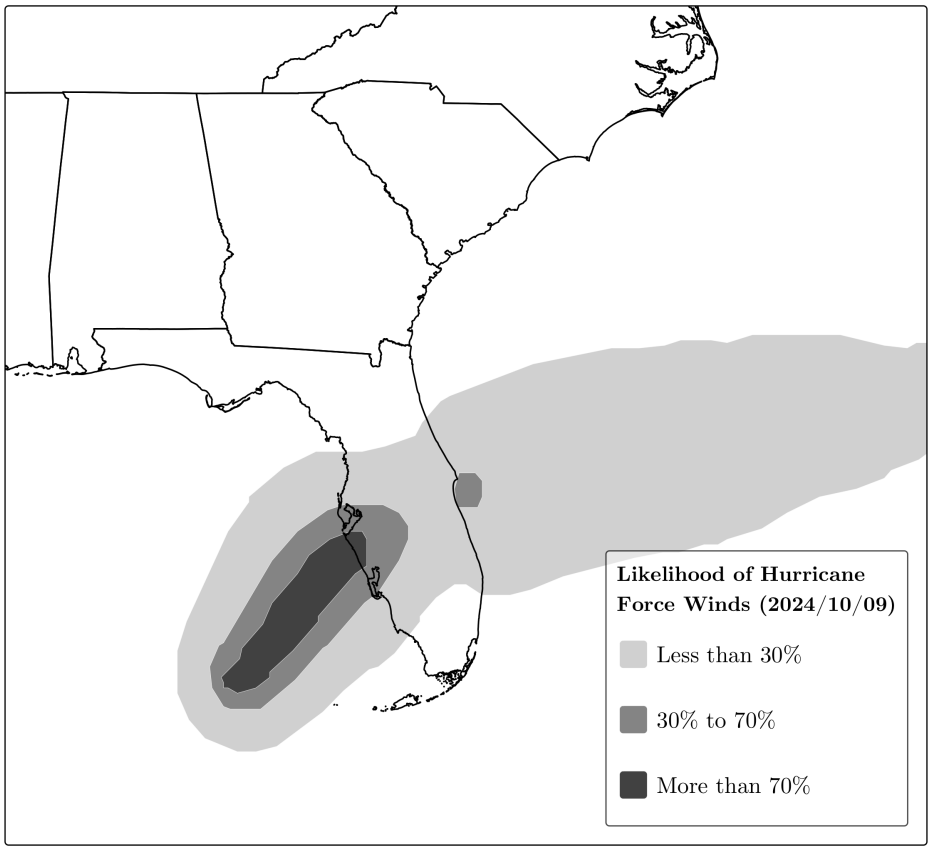


Figure 6: Hurricane Force Wind Probabilities

Notes: This figure presents hurricane-force wind speed probabilities for Hurricane Milton as provided by the National Hurricane Center (NHC) on October 9, 2024, at 11:00AM.

forecasted exposed populations in terms of characteristics related to vulnerability—e.g., low income, elderly, non-White populations.

This capability was utilized in advance of Hurricane Milton (USCB, 2024a). We combined the gridded EIF with the forecast products that the National Hurricane Center produced immediately before Hurricane Milton made landfall in Florida on October 9th. These forecasts include predicted probabilities of hurricane-force and tropical-storm-force winds, as well as the predicted probability of flash flooding. Figure 6 shows the extent of the forecasts for hurricane force winds as of 11:00am on the day Milton made landfall.

We intersected the version of these forecast products produced at 11:00am ET on the day Milton made landfall with the gridded EIF. This allowed usto explore the characteristics of the population that was forecast to be exposed to environmental hazards associated with Hurricane Milton. Table 1 summarizes the demographic characteristics of the population that was forecasted to be exposed to hurricane force winds, compared to the demographics of the overall real-time gridded EIF for 2024. The population forecasted to be exposed to hurricane force winds was substantially older and more likely to be non-Hispanic White relative to the overall U.S. population. The Gridded EIF also allowed usto explore more detailed demographic categories, such as the population in each race-by-income decile forecasted to be exposed to hurricane-related hazards, as reported in USCB (2024a). After landfall we intersected the Gridded EIF with post-landfall geospatial information about the realization of hazards to generate information on the population actually exposed. This was then combined with the pre-landfall analysis to explore the characteristics of the population exposed to forecast errors. The complete analysis can be found in USCB (2024b).

Pre- and post-landfall analysis of hurricanes is only one such application of the gridded EIF for real time analysis purposes. There are several other applications under active development. Geospatial data and forecast products for a much broader set of discrete extreme weather events beyond exposure to hurricane wind fields can be combined with the gridded EIF to explore a much broader array of hazards. This includes extreme precipitation events,

coastal flooding due to storm surge, riverine flooding, extreme heat events, tornadoes and other convective storms. Finally, by combining the gridded EIF with the National Weather Service's Air Quality forecast product and real-time pollution monitoring data, it is possible to track the characteristics of the population exposed to hourly or daily fluctuations in air quality, analysis that could be embedded in a continuous monitoring tool.

Table 1: Socioeconomic characteristics of the population forecasted to experience hurricane-force winds (70 percent or more probability) compared to the national population.

	Population	Percent	National percent	Difference
Non-Hispanic White	547,242	70.71	55.03	15.68
Non-Hispanic Black	44,713	5.78	11.60	-5.82
Hispanic	84,998	10.98	15.80	-4.82
Non-Hispanic Asian	12,814	1.66	4.13	-2.47
Non-Hispanic AIAN	1,678	0.22	0.83	-0.61
Other	82,515	10.66	12.61	-1.95
Over the Age of 65	244,158	31.55	18.26	13.29
Under the Age of 18	109,691	14.17	19.69	-5.52
Lower Income	154,601	19.95	20.02	-0.07
Total Population Exposed	773,960			

Notes: This table presents the sociodemographic characteristics of the populations forecasted to be exposed to hurricane-force winds with probability greater than 70 percent, using probabilities provided by the National Hurricane Center (NHC) on October 9, 2024, at 11:00AM. Column 1 provides population counts. Column 2 provides percentages of the total exposed population. Column 3 reports each group's share of the national population. Column 4 reports the difference in group shares. Source: Gridded Environmental Impacts Frame and NHC data.

4 Access to the Gridded EIF

The Gridded EIF will be made available to the public, facilitating exploration of the distribution of exposure to environmental hazards. These files will be distributed via the Census Bureau's website. For each year 1999–2023 and for the "real time" 2024, race-age-sex and race-income datasets will be made available, indexed by grid point centroid. These will be distributed in Parquet format due to the large size of the gridded datasets. Additionally, the harmonized pollution data described above will be made available for the years 1999-2022, again indexed by grid point centroid. To conduct geospatial analyses of other environmental hazards, or indeed any spatial variation, we will also distribute a grid topology file in R data format that provides a mapping to the gridded EIF. Example code to combine the gridded EIF files will be provided in additional documentation.

The Gridded EIF provides an easily accessible way for researchers to engage with many of the patterns and trends available in the restricted use microdata, without the access restrictions and disclosure constraints that come with the use of confidential Census Bureau data. As needs and uses for this gridded data evolve, the Census Bureau will continue to update and enhance the information available in the gridded EIF.

References

Abowd, John, Robert Ashmead, Ryan Cumings-Menon, Simson Garfinkel, Micah Heineck, Christine Heiss, Robert Johns, Daniel Kifer, Philip Leclerc, Ashwin Machanavajjhala, Brett Moran, William Sexton, Matthew Spence, and Pavel Zhuravlev, "The 2020 Census Disclosure Avoidance System TopDown Algorithm," *Census Working Paper Series*, 2022.

Canonne, Clément L., Gautam Kamath, and Thomas Steinke, "The Discrete Gaussian for Differential Privacy," *CoRR*, 2020, *abs/2004.00010.*

Chetty, Raj and John Friedman, "A Practical Method to Reduce Privacy Loss when Disclosing Statistics Based on Small Samples," *NBER Working Paper Series*, 2019.

Colmer, Jonathan, Suvy Qin, John Voorheis, and Reed Walker, "Income, Wealth, and Environmental Inequality in the United States," *Working Paper*, 2024.

Di, Qian, Heresh Amini, Liuhua Shi, Itai Kloog, Rachel Silvern, James Kelly, M Benjamin Sabath, Christine Choirat, Petros Koutrakis, Alexei Lyapustin, Yujie Wang, Loretta J Mickley, and Joel Schwartz, "An Ensemble-based Model of PM2.5 Concentration Across the Contiguous United States with High Spatiotemporal Resolution," *Environment International*, 2019, *130.*

_ , _ , _ , _ , _ , _ , M. Benjamin Sabath, Christine Choirat, Petros Koutrakis, Alexei Lyapustin, Yujie Wang, Loretta J. Mickley, and Joel Schwartz, "Assessing NO2 Concentration and Model Uncertainty with High Spatiotemporal Resolution across the Contiguous United States," *Environmental Science Technology*, 2020, *54.*

Kim, Sun-Young, Matthew Bechle, Steve Hankey, Lianne Sheppard, Adam A. Szpiro, and Julian D. Marshall, "Concentrations of criteria pollutants in the contiguous U.S., 1979 – 2015: Role of prediction model parsimony in integrated empirical geographic regression," *PLoS ONE*, 2020, *15.*

Kondo, Illenin, Kevin Rinz, Natalie Gubbay, Brandon Hawkins, John Voorheis, and Abigail Wozniak, "Granular Income Inequality and Mobility Using IDDA: Exploring Patterns across Race and Ethnicity," *NBER Working Paper*, 2024.

Requia, Weeberb, Qian Di, Rachel Silvern, James T. Kelly, Petros Koutrakis, Loretta J. Mickley, Melissa P. Sulprizio, Heresh Amini, Liuhua Shi, and Joel Schwartz, "An Ensemble Learning Approach for Estimating High Spatiotemporal Resolution of Ground-level Ozone in the Contiguous U.S.," *Environmental Science Technology*, 2020, *54.*

Shen, Siyuan, Chi Li, Aaron van Donkelaar, Nathan Jacobs, Chenguang Wang, and Randall V. Martin, "Enhancing Global Estimation of Fine Particulate Matter Concentrations by Including Geophysical a Priori Information in Deep Learning," *ACS EST Air*, 2024, *1.*

USCB, "Real-Time Analysis: Hurricane Milton," *US Census Bureau Technical Report*, October 9th 2024.

_ , "Real-Time Analysis: Hurricane Milton, Post-Landfall Report," *US Census Bureau Technical Report*, October 28th 2024.

van Donkelaar, Aaron, Melanie S. Hammer, Liam Bindle, Michael Brauer, Jeffery R. Brook, Michael J. Garay, N. Christina Hsu, Olga V. Kalashnikova, Ralph A. Kahn, Colin Lee, Robert C. Levy, Alexei Lyapustin, Andrew M. Sayer, and Randall V. Martin, "Monthly Global Estimates of Fine Particulate Matter and Their Uncertainty," *Environmental Science Technology*, 2021, *55*.

Voorheis, John, Jonathan Colmer, Kendall Houghton, Eva Lyubich, Mary Munro, Cameron Scalera, and Jennifer Withrow, "Building the Prototype Census Environmental Impacts Frame," *CES Working Paper CES-23-20*, 2023.